

Random Hyperbolic Surface Spectral Rigidity — cycles 7-9

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Contents

Random Hyperbolic Surface Spectral Rigidity — cycles 7-9	1
Abstract	1
Introduction	1
Approach	2
Source Inventory and Timeline	2
Findings	7
Discussion	9
Open Questions	9
References	10
Appendix: Implementation Details	10

Random Hyperbolic Surface Spectral Rigidity — cycles 7-9

Abstract

Cycles 7-9 continued M3-computational-probes, the computational milestone for the long-exposure study of Kim and Tao’s paper Eigenvalue rigidity of hyperbolic surfaces in the random cover model. Earlier cycles had already validated the proof map and proof ledger, including the role of polynomial-method common-fixed-point estimates and the diagonal subtraction term in the eigenfunction delocalization proof. These three cycles turned that proof-ledger understanding into a more structured computational ladder.

Cycle 7 added folded trajectory quotient profiles to the Cycle 6 random-permutation fixed-point probe. This separated cyclic/rank-one word families from rank-two/noncyclic controls structurally, but it also confirmed that fixed-basepoint Monte Carlo remains too sparse for rank-two eight-word families.

Cycle 8 changed the observable from fixed-basepoint intersections to direct injective labelled-graph embedding counts. This solved the sparsity problem at the toy level: the cyclic eight-edge template remained order one, while the rank-two eight-edge template had stable nonzero estimates at the expected n^{-1} scale. The auditor fixed one inverse-label estimator bug; the published lowercase-template numerical conclusions were unchanged.

Cycle 9 used the Cycle 8 benchmark pair for polynomial-window diagnostics in $x = 1/n$. Degree-3 Chebyshev-window fits extrapolated accurately toward $x = 0$, while deliberately underdetermined degree-6 and degree-8 fits showed large derivative and coefficient growth. The auditor validated this as a toy Markov-amplification diagnostic, not as the actual MPvH/MP23 polynomial or a hyperbolic trace statistic.

Introduction

The campaign’s first two milestones established the local proof architecture of the paper. In that architecture, random-cover trace and pre-trace estimates are reduced to finite permu-

tation statistics. The proof then controls polynomial expressions in reciprocal cover degree, with Markov-type derivative bounds producing visible exponent losses.

Cycles 7-9 focused on a narrow computational question: how much of that finite permutation mechanism can be reproduced in simple random-permutation toy models before adding hyperbolic length weights or actual random-cover spectra?

The answer from these cycles is a three-step computational ladder:

1. A folded trajectory quotient classifies word families by cyclic/rank-one versus rank-two/noncyclic structure.
2. Direct labelled-graph embedding counts give stable nonzero estimates for eight-word rank-two toy templates.
3. Polynomial fits to normalized embedding counts expose a reproducible interpolation-stability cost.

These are validated M3 slices, not closure of M3. The next recommended M3 target remains a Schreier/random-cover toy spectral probe using the Cycle 9 benchmark as a reference.

Approach

Cycle 7 introduced a **folded trajectory quotient**. For a tuple of words, the script traces each reduced word from a common symbolic basepoint, identifies terminal points back to that basepoint, merges identical labelled edges, and records graph invariants. This is a simplified labelled trajectory quotient, not full Stallings folding and not the MPvH/MP23 quotient expansion.

Cycle 8 introduced a **labelled-graph embedding count**. For a small labelled directed graph H , the observable counts injective vertex maps into $[n]$ such that every labelled edge constraint $\text{sigma_label}(f(u)) = f(v)$ is satisfied by independent random permutations. This moves closer to quotient-graph embeddings than asking how many points are fixed by all words in a tuple.

Cycle 9 introduced a **polynomial-window diagnostic**. It reads the Cycle 8 normalized embedding counts as values of a function of $x = 1/n$, fits polynomial surrogates in a Chebyshev basis, and records hold-out error, extrapolation error near $x = 0$, derivative at zero, and coefficient norms.

Source Inventory and Timeline

Cycle 7: Folded Word-Graph Probe

Researcher session 980c616d-f011-4fba-a76f-a01ddad32997 set the task: preserve the Cycle 6 random-permutation harness but add quotient-graph structure so cyclic diagonal families and rank-two/noncyclic families could be distinguished before sampling.

Worker session 408bb527-400a-4300-844c-9fa8a3e0f2f7 produced:

- scripts/probe_folded_word_graphs.py
- tests/test_folded_word_graphs.py
- data/polynomial_method/folded_word_graph_probe.csv
- data/polynomial_method/folded_word_graph_summary.csv
- reports/computational_probes/m3_folded_word_graph_probe.md
- reports/figures/m3_folded_graph_rank_scaling.png
- reports/figures/m3_folded_graph_profile_heatmap.png

The final run wrote 52,000 raw rows across 13 families and $n = 50, 100, 200, 400$.

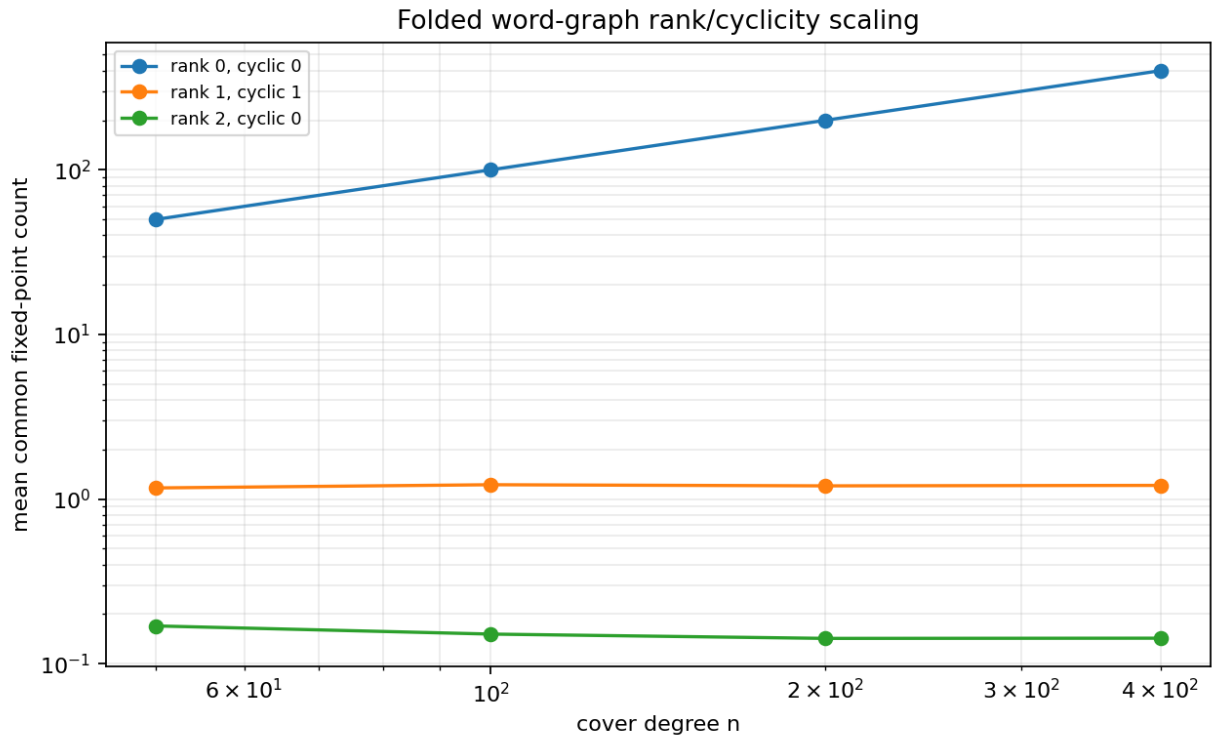


Figure 1: Common-fixed-point scaling grouped by folded word-graph rank/cyclicity.

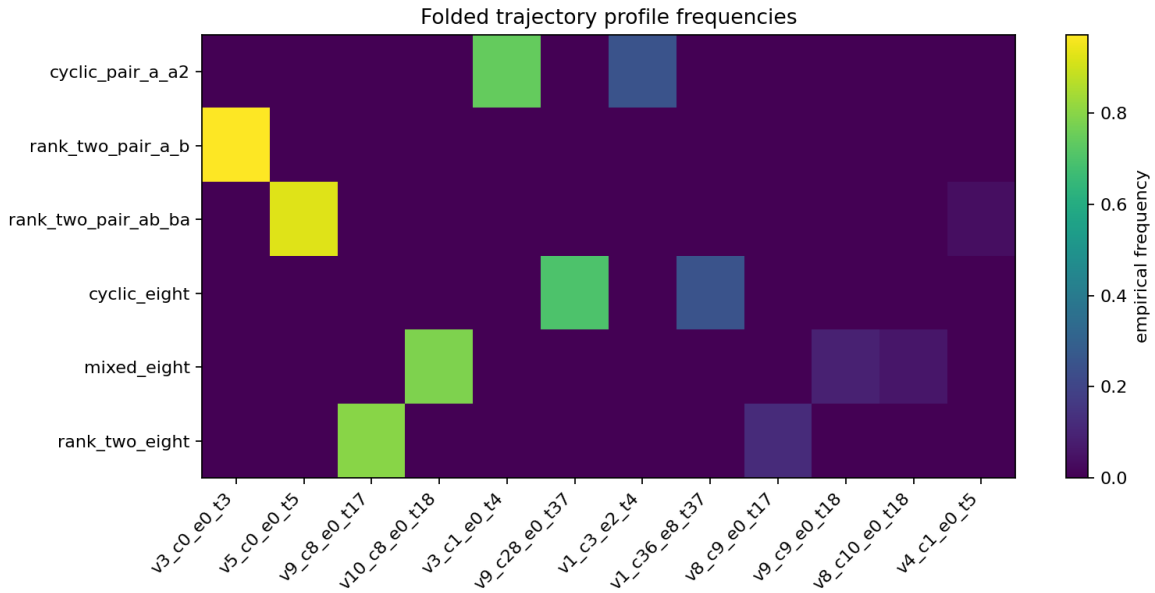


Figure 2: Empirical frequency of folded trajectory profiles across cyclic, mixed, and rank-two word families.

Auditor session 5418764b-feb6-4e41-8583-2e5e0f906c83 validated the slice. The auditor found no critical or moderate issues, confirmed tests and figures, and kept M3-computational-probes as in-progress.

Cycle 8: Labelled Graph Embedding Probe

Researcher session ab5df6ec-9450-48c8-a747-1f6dc5e2f9d0 directed the next step: replace fixed-basepoint intersections with direct embeddings of small labelled quotient graphs.

Worker session 9d05ccd8-154c-446b-b477-28ccab54d6e9 produced:

- scripts/probe_labelled_graph_embeddings.py
- tests/test_labelled_graph_embeddings.py
- data/polynomial_method/labelled_graph_embedding_probe.csv
- data/polynomial_method/labelled_graph_embedding_summary.csv
- reports/computational_probes/m3_labelled_graph_embedding_probe.md
- reports/figures/m3_labelled_embedding_scaling.png
- reports/figures/m3_labelled_embedding_normalized.png

The final data contain 80 rows over eight templates and the n grid 3, 4, 8, 20, 50, 100, 200, 400.

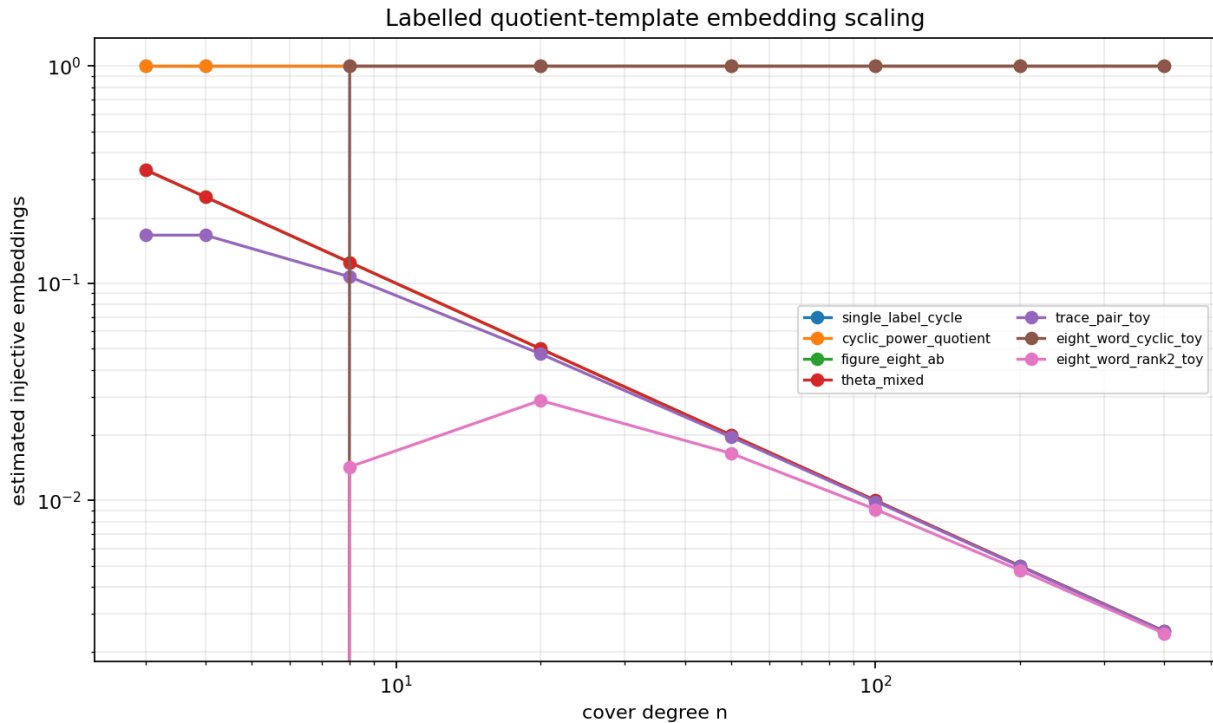


Figure 3: Estimated labelled-graph embedding counts versus n , grouped by quotient-template rank.

Auditor session 9474882f-ed40-4e43-806a-b21e8936e24c found one moderate implementation defect: inverse-labelled edges were mishandled in the Monte Carlo expectation estimator. The auditor repaired the issue in scripts/probe_labelled_graph_embeddings.py, added a regression test, and appended ledger event 1b6b42a7-55c3-4d82-9c54-7cf3f7a698e2. The final Cycle 8 templates used lowercase labels only, so the reported numerical results were unaffected.

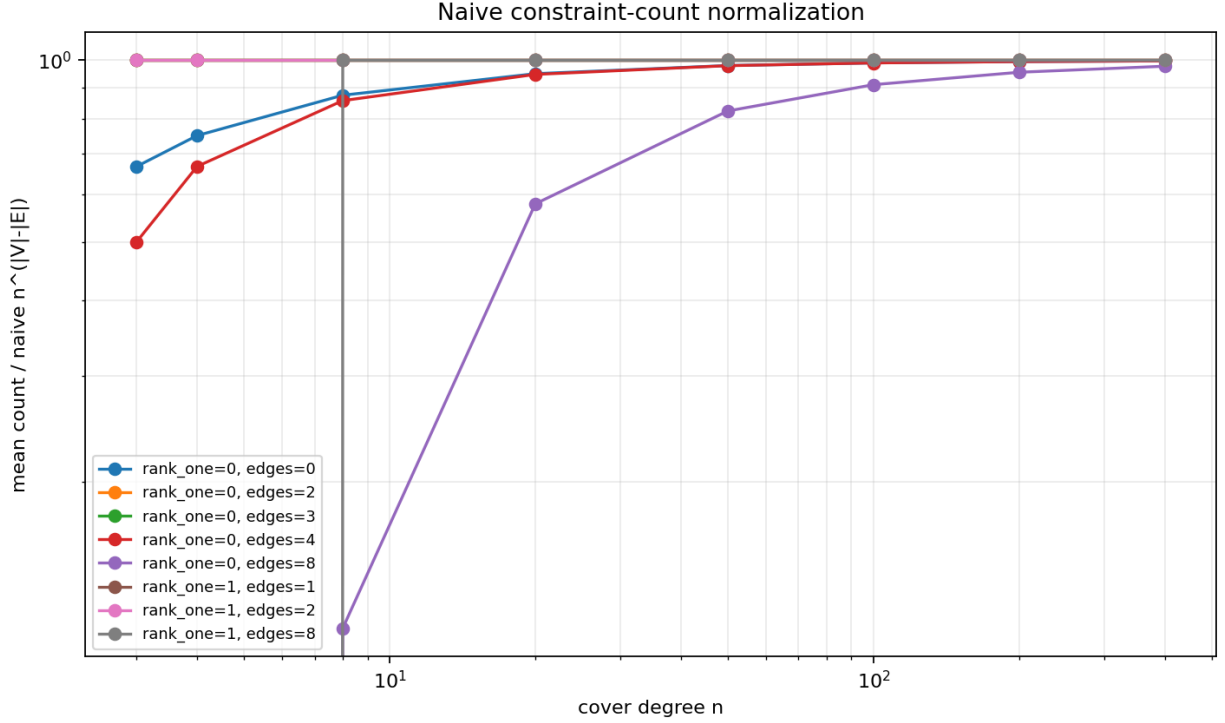


Figure 4: Embedding counts normalized by the naive $n^{|V|-|E|}$ constraint-count scale.

Cycle 9: Polynomial-Window Diagnostics

Researcher session 906e78cd-e243-41f7-ac07-c439cd89eb3f directed a polynomial-window experiment using the validated Cycle 8 benchmark pair. The goal was to probe interpolation and Markov-type derivative amplification in a toy setting.

Worker session 0ef0e4ce-f292-4584-acd3-b25c7c6a678d produced:

- scripts/probe_polynomial_window_diagnostics.py
- tests/test_polynomial_window_diagnostics.py
- data/polynomial_method/polynomial_window_diagnostics.csv
- data/polynomial_method/polynomial_window_fit_summary.csv
- reports/computational_probes/m3_polynomial_window_diagnostics.md
- reports/figures/m3_polynomial_window_fit_error.png
- reports/figures/m3_polynomial_window_derivative_growth.png
- reports/figures/m3_polynomial_window_extrapolation.png

The final run wrote 7,344 diagnostic rows and 18 fit-summary rows.

Auditor session ea490aa2-d83a-4f19-994a-85aedb667f55 validated the slice. The audit confirmed that tests passed, smoke output was generated, all figures were readable at 1440 x 864 RGBA, and repository checks passed with known warnings only.

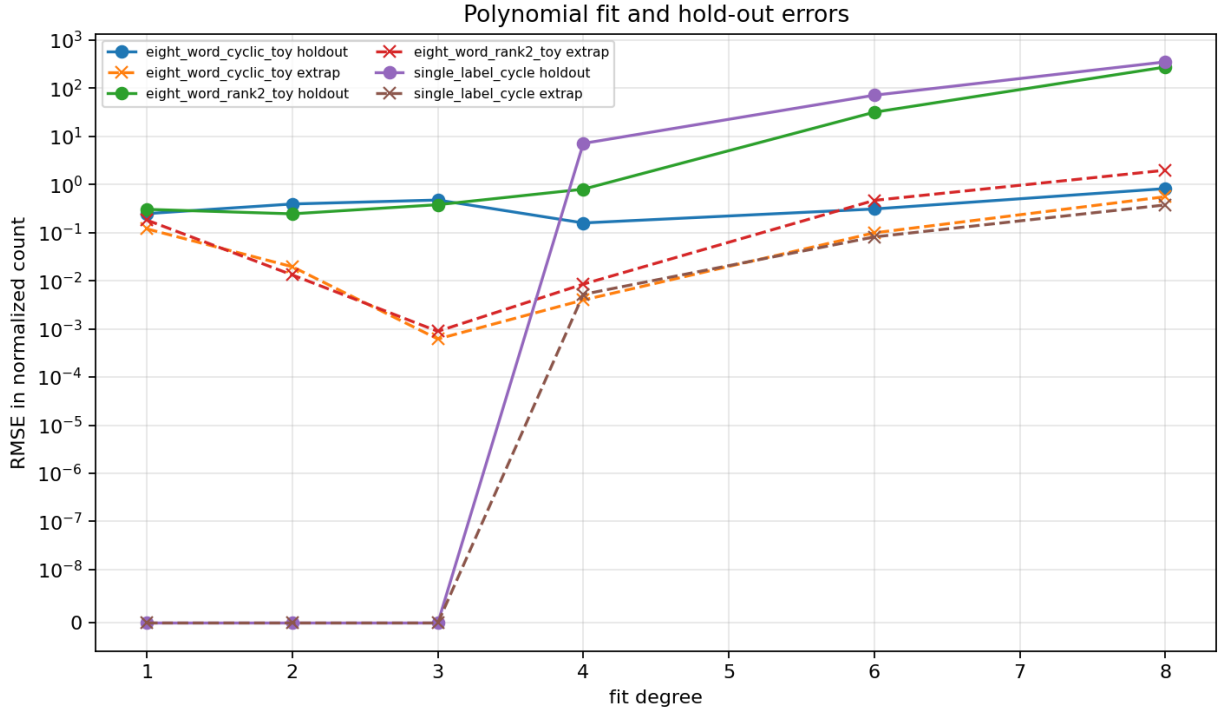


Figure 5: Polynomial fit and hold-out errors for cyclic and rank-two labelled-embedding benchmark templates.

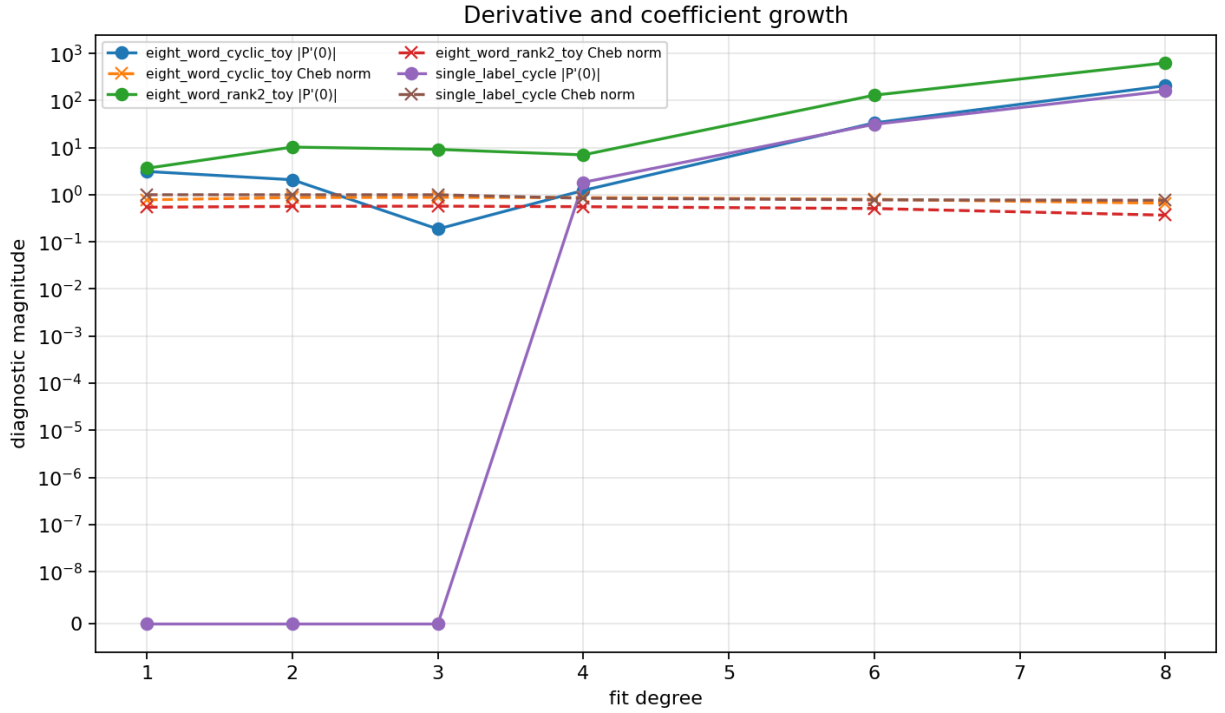


Figure 6: Derivative and coefficient growth versus polynomial degree, illustrating the toy Markov-amplification mechanism.

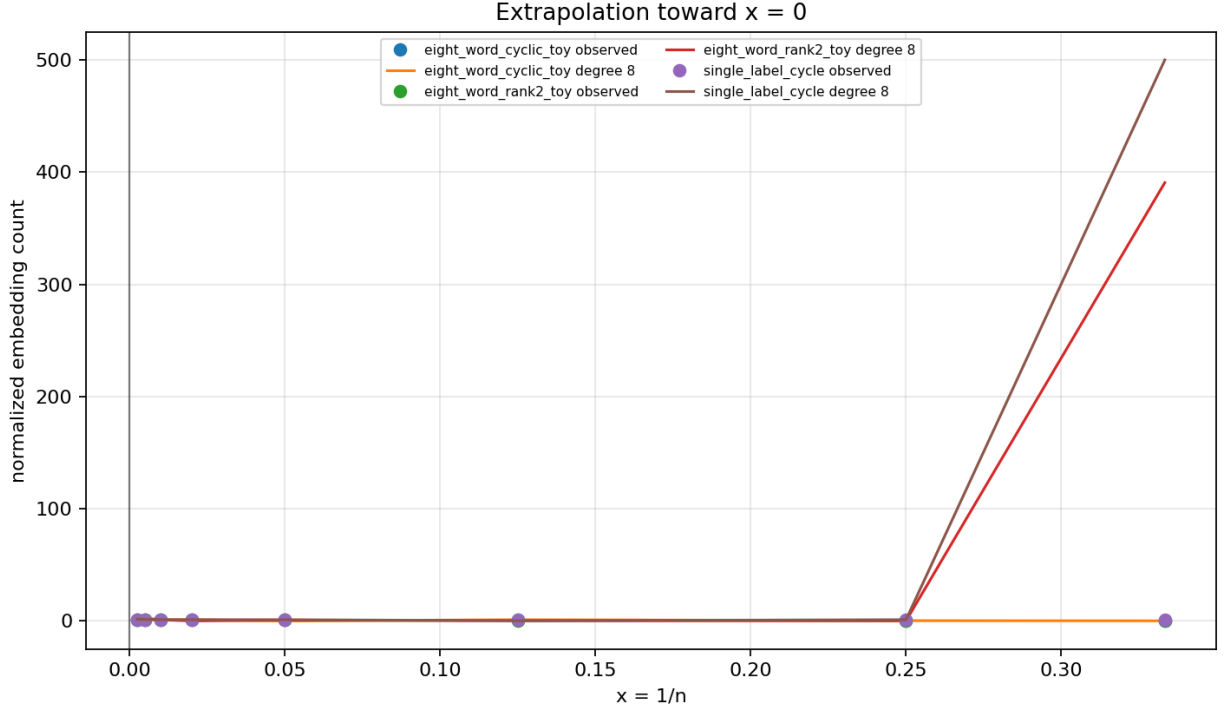


Figure 7: Extrapolation toward $x = 0$ for normalized embedding-count observables.

Findings

Finding 1: Folded quotient profiles explain the Cycle 6 null result

Cycle 6 showed that naive pointwise common fixed sets are too crude for multiword rank-two families: adding composite words can collapse to the same fixed-point constraint as a smaller pair.

Cycle 7 made this null result structural. The folded trajectory quotient classified `cyclic_pair_a_a2` and `cyclic_eight` as generator-rank-one cyclic families, and both had mean common fixed-point counts near one. Rank-two/noncyclic families were classified separately and had much smaller means.

Selected Cycle 7 means at $n = 400$:

family	generator rank	cyclic flag	mean
<code>cyclic_pair_a_a2</code>	1	1	1.016
<code>cyclic_eight</code>	1	1	1.016
<code>rank_two_pair_a_b</code>	2	0	0.000
<code>rank_two_pair_ab_ba</code>	2	0	0.006
<code>mixed_eight</code>	2	0	0.000
<code>rank_two_eight</code>	2	0	0.000

The conclusion was not that fixed-basepoint Monte Carlo estimates the eight-word mechanism well. It was that quotient structure should be tracked explicitly before choosing an estimator.

Finding 2: Direct labelled-graph embeddings solve the eight-word sparsity problem at toy level

Cycle 8 changed the observable. Instead of sampling basepoints fixed by all words, it estimated expected injective labelled-graph embeddings. This made rank-two eight-word templates measurable.

Selected Cycle 8 estimates at $n = 400$:

template	count estimate	naive power	normalized count
eight_word_cyclic_toy	1.000000	0	1.000000
eight_word_rank2_toy	0.00244389	-1	0.977557
figure_eight_ab	0.0025	-1	1.000000
trace_pair_toy	0.00249373	-1	0.997494

The rank-two eight-edge template no longer vanished under sampling. Its raw count scaled like n^{-1} , while normalization by the naive constraint scale $n^{|V|-|E|}$ brought it close to one.

This supports the Cycle 8 interpretation: the main cyclic/rank-two separation in the toy embedding model is constraint dimension, not an extra anomalous constant after normalization.

Finding 3: The Cycle 8 audit repaired inverse-label semantics

The Cycle 8 auditor found that uppercase inverse-labelled edges were supported in exact realized checks but mishandled in the Monte Carlo expectation estimator. The repair normalized uppercase labels as reversed partial-permutation constraints.

The published templates used lowercase labels, so the numerical conclusions remained aligned with the worker report. The repair matters for future work because later quotient graphs may naturally contain inverse-labelled edges.

Finding 4: Degree-3 polynomial-window fits are the current benchmark

Cycle 9 fit normalized Cycle 8 embedding counts as functions of $x = 1/n$. The recommended benchmark is:

- templates: eight_word_cyclic_toy and eight_word_rank2_toy
- basis: Chebyshev-window fitting
- degree: 3
- use degree 6 and 8 only as instability stress tests

Selected Cycle 9 fit-summary values:

template	degree	holdout RMSE	extrapolation RMSE	derivative at 0	coefficient norm
single_label_cycle	1	2.22e-16	2.22e-16	-3.30e-16	1
eight_word_cyclic_toy	3	4.76e-1	6.17e-4	-1.87e-1	47.96

template	degree	holdout RMSE	extrapolation RMSE	derivative at 0	coefficient norm
eight_ word_ rank2_toy	3	3.81e-1	8.90e-4	-9.19	35.71
eight_ word_ cyclic_toy	8	8.23e-1	5.58e-1	2.06e2	1.02e8
eight_ word_ rank2_toy	8	2.76e2	1.98	-6.29e2	7.24e8

The exact `single_label_cycle` control was recovered to machine precision at low degrees. Its high-degree failures were intentionally treated as conditioning stress tests, not as failures of the constant sequence.

Finding 5: The toy Markov-amplification signal is concrete but limited

Cycle 9 gives a concrete analogue of the proof ledger’s Markov-loss mechanism: as fit degree increases in an underdetermined window, derivative and coefficient diagnostics grow sharply.

The auditor accepted this conclusion with a clear limitation. The diagnostic measures interpolation conditioning for normalized expected labelled-embedding counts. It is not the actual MPvH/MP23 polynomial, and it is not a hyperbolic trace statistic.

Discussion

Cycles 7-9 advanced M3 from a raw random-permutation baseline into a structured computational framework. Each cycle addressed a defect exposed by the previous one.

Cycle 7 explained why raw common fixed-point intersections were inadequate for rank-two eight-word families. Cycle 8 replaced that sparse observable with direct labelled-graph embeddings. Cycle 9 then used the stable Cycle 8 embedding data to test interpolation stability.

The resulting ladder is now useful for future work because it separates three issues that were previously entangled:

- **Structure:** cyclic/rank-one versus rank-two/noncyclic quotient type.
- **Scale:** raw embedding-count suppression governed by $n^{\{|V| - |E|\}}$.
- **Interpolation:** stability or instability of polynomial fits near $x = 0$.

The strongest next target, recorded by the Cycle 9 auditor, is a Schreier/random-cover toy spectral probe. That would move M3 closer to the paper’s geometric setting while preserving the degree-3 cyclic/rank-two polynomial-window benchmark as a reference.

Open Questions

1. How should the current labelled-graph templates be embedded into an actual Schreier/random-cover toy spectral model without losing the clean cyclic/rank-two benchmark?
2. Can the degree-3 polynomial-window benchmark be made less toy-like by adding hyperbolic length weights or trace-window weights while preserving exact controls?

3. Which part of the paper's Markov loss is best represented by the current derivative/coefficient diagnostics: trace-side $q^{\{2 \text{ kappa}\}}$, pre-trace fourth-moment $q^{\{4 \text{ kappa}\}}$, or only the general interpolation-conditioning phenomenon?
4. Can inverse-labelled quotient templates now be added safely after the Cycle 8 audit repair, and do they change the normalized embedding-count behavior?
5. At what point should M3 close? It now has four validated slices, but the audit guidance keeps M3 open for at least a Schreier/random-cover toy spectral probe.

References

No REFERENCES.md file was present in the workspace during this reporter pass. As in the cycles 4-6 report, this section therefore lists local sources and session records rather than global numbered references.

- Kim and Tao, Eigenvalue rigidity of hyperbolic surfaces in the random cover model, local files 2603.01127.pdf and 2603.01127.txt.
- Cycle 7 researcher session 980c616d-f011-4fba-a76f-a01ddad32997.
- Cycle 7 worker session 408bb527-400a-4300-844c-9fa8a3e0f2f7.
- Cycle 7 auditor session 5418764b-feb6-4e41-8583-2e5e0f906c83.
- Cycle 8 researcher session ab5df6ec-9450-48c8-a747-1f6dc5e2f9d0.
- Cycle 8 worker session 9d05ccd8-154c-446b-b477-28ccab54d6e9.
- Cycle 8 auditor session 9474882f-ed40-4e43-806a-b21e8936e24c.
- Cycle 9 researcher session 906e78cd-e243-41f7-ac07-c439cd89eb3f.
- Cycle 9 worker session 0ef0e4ce-f292-4584-acd3-b25c7c6a678d.
- Cycle 9 auditor session ea490aa2-d83a-4f19-994a-85aedb667f55.
- reports/computational_probes/m3_folded_word_graph_probe.md.
- reports/computational_probes/m3_labelled_graph_embedding_probe.md.
- reports/computational_probes/m3_polynomial_window_diagnostics.md.

Appendix: Implementation Details

Code Organization

Cycle 7 files:

- scripts/probe_folded_word_graphs.py: 398 lines.
- tests/test_folded_word_graphs.py: 59 lines.
- reports/computational_probes/m3_folded_word_graph_probe.md: 93 lines.
- data/polynomial_method/folded_word_graph_probe.csv: 52,001 lines including header.
- data/polynomial_method/folded_word_graph_summary.csv: 53 lines including header.

Cycle 8 files:

- scripts/probe_labelled_graph_embeddings.py: 444 lines.
- tests/test_labelled_graph_embeddings.py: 87 lines.
- reports/computational_probes/m3_labelled_graph_embedding_probe.md: 97 lines.
- data/polynomial_method/labelled_graph_embedding_probe.csv: 81 lines including header.
- data/polynomial_method/labelled_graph_embedding_summary.csv: 81 lines including header.

Cycle 9 files:

- scripts/probe_polynomial_window_diagnostics.py: 320 lines.
- tests/test_polynomial_window_diagnostics.py: 72 lines.
- reports/computational_probes/m3_polynomial_window_diagnostics.md: 91 lines.

- data/polynomial_method/polynomial_window_diagnostics.csv: 7,345 lines including header.
- data/polynomial_method/polynomial_window_fit_summary.csv: 19 lines including header.

Validation Results

Cycle 7 validation, from session 5418764b-feb6-4e41-8583-2e5e0f906c83:

- py_compile: passed.
- Direct tests: passed.
- Smoke run: passed.
- Final CSV: 52,000 rows, 13 families, four n values, 1,000 rows per (n, family).
- Figures: readable, 1280 x 800 RGBA and 1600 x 800 RGBA.
- Decision: VALIDATED.

Cycle 8 validation, from session 9474882f-ed40-4e43-806a-b21e8936e24c:

- py_compile: passed after audit repair.
- Direct tests: passed after audit repair.
- Smoke run: passed.
- Inverse-label regression: exact and estimator outputs agreed within roundoff.
- Final CSV: 80 rows, eight templates, modes exact and monte_carlo.
- Figures: readable, both 1440 x 880 RGBA.
- Decision: VALIDATED.

Cycle 9 validation, from session ea490aa2-d83a-4f19-994a-85aedb667f55:

- py_compile: passed.
- Direct tests: passed.
- Smoke run: passed, writing 252 diagnostic rows and 9 fit rows.
- Final CSVs: 7,344 diagnostic rows and 18 fit-summary rows.
- Figures: readable, all three 1440 x 864 RGBA.
- Decision: VALIDATED.

Reporter-Side Checks

After updating MANIFEST.md, reporter-side checks were run:

- python3 -m long_exposure.tools.promise_check .: passed with warnings only. It reported 25 ledger events and known warnings for noncanonical docs/paper_map/, pending M4/M5/M6 milestones, and orphan prior periodic reports.
- python3 -m long_exposure.tools.org_check .: passed with warnings only. It reported known root-file warnings and historical docs/ figure-location warnings.

Manifest Update

MANIFEST.md was replaced as a current snapshot. No ## Key Files section existed, so no final-reporter-owned section needed preservation.

Current manifest totals:

Metric	Value
Campaign scripts	5
Campaign script lines	1,504
Campaign test files	4
Campaign test lines	275

Metric	Value
Markdown proof/map/report artifacts	20
PNG figures	17
Canonical CSV datasets	9
Promise ledger events	25

Session Cross-Reference Map

Cycle	Role	Session ID	Role in report
7	researcher	980c616d-f011-4fba-a76f-a01ddad32997	Defined folded word-graph/common-quotient task.
7	worker	408bb527-400a-4300-844c-9fa8a3e0f2f7	Built folded trajectory quotient probe and outputs.
7	auditor	5418764b-feb6-4e41-8583-2e5e0f906c83	Validated Cycle 7 slice.
8	researcher	ab5df6ec-9450-48c8-a747-1f6dc5e2f9d0	Defined direct labelled-graph embedding task.
8	worker	9d05ccd8-154c-446b-b477-28ccab54d6e9	Built embedding-count probe and outputs.
8	auditor	9474882f-ed40-4e43-806a-b21e8936e24c	Repaired inverse-label estimator and validated Cycle 8.
9	researcher	906e78cd-e243-41f7-ac07-c439cd89eb3f	Defined polynomial-window diagnostic task.
9	worker	0ef0e4ce-f292-4584-acd3-b25c7c6a678d	Built polynomial-window diagnostic suite and outputs.
9	auditor	ea490aa2-d83a-4f19-994a-85aedb667f55	Validated Cycle 9 slice.